Edge AI-Driven Stealth Drone for Autonomous Object Following with Zero Trust Command Security

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***Abstract*—Unmanned Aerial Vehicles (UAVs) have gained significant traction in surveillance and assistive applications, yet most low-cost drones lack intelligent perception, autonomous tracking, and secure communication. This paper presents *SecureDrone*, a Secure Vision-Based Human Tracking and Autonomous Drone Control System using Edge Artificial Intelligence (Edge AI). The system employs a Raspberry Pi 5 with a CSI camera module to perform real-time human detection using the YOLOv8 deep learning model. By offloading all vision processing to an external edge device, the computational burden on the lightweight ESP32-based LiteWing drone is eliminated, preserving flight endurance and stability. Detected human positions are continuously analyzed through bounding box centroid deviation and proximity estimation to generate autonomous movement commands. A token-based Zero Trust authentication mechanism secures the wireless command channel, preventing unauthorized access, spoofing, and replay attacks. Experimental evaluation in controlled indoor environments demonstrates stable human-following with an average end-to-end latency of 230–350 ms, detection rates of 8–12 FPS on the Raspberry Pi 5, a mean detection confidence of 0.87, and consistent tracking accuracy under varying lighting conditions. The modular Sense–Analyze–Act architecture supports future integration of obstacle avoidance, SLAM-based navigation, and advanced tracking algorithms.**

***Index Terms*—Edge AI, YOLOv8, Autonomous Drone, Human Tracking, Zero Trust Security, ESP32, Raspberry Pi 5, Visual Servoing, UAV, Real-Time Object Detection**

I. Introduction

The operational paradigm of unmanned aerial vehicles (UAVs) has shifted from manual remote control toward proactive, edge-intelligent autonomy. Traditional drone systems rely on line-of-sight operation, predefined flight paths, and human judgment, constraining their scalability in dynamic environments such as smart surveillance, disaster response, and assistive robotics [15]. The integration of Edge Artificial Intelligence (Edge AI) enables on-device data processing, reducing latency, improving responsiveness, and enhancing privacy by minimizing data transmission to remote servers [10].

A critical gap exists in the autonomous drone literature: the systematic exclusion of low-cost, resource-constrained drones from intelligent vision-based control. Hardware categories most widely deployed—affordable consumer drones, DIY kits, and educational platforms—are precisely those least capable of running deep learning inference onboard. Mounting GPU-class accelerators such as NVIDIA Jetson Nano onto lightweight platforms like the ESP32-based LiteWing drone exceeds thrust capacity, drains batteries within minutes, and introduces vibration-induced inference errors [13]. Onboard processing on microcontroller-class hardware introduces 10–30 seconds of detection latency per frame, far exceeding acceptable thresholds for tracking a moving human subject.

Cloud-based inference pipelines present an alternative but introduce round-trip latencies of 200–800 ms per frame [14], which is unacceptable for real-time human following where positional decisions must occur within tens of milliseconds. Furthermore, wireless drone communication channels are inherently vulnerable to signal spoofing, command injection, and replay attacks [16]. Most low-cost platforms lack cryptographic verification capabilities, creating open channels for unauthorized control. According to the Federal Aviation Administration, over 1.5 million drones were registered in the United States alone by 2023, with unauthorized drone incidents near sensitive facilities increasing by over 300% in the past five years [5].

This paper presents **SecureDrone**, a system that addresses these challenges through three key contributions:

1) **Edge-Offloaded Vision Pipeline:** All YOLOv8 inference is performed on a Raspberry Pi 5, keeping the LiteWing drone lightweight while preserving full autonomous intelligence at 8–12 FPS.
2) **Dual-Layer Visual Servoing:** A hierarchical control strategy combining bounding box centroid deviation analysis for directional commands and proximity estimation via bounding box scaling for adaptive distance control.
3) **Zero Trust Command Authentication:** A token-based verification mechanism where every command is validated for structural integrity, parameter ranges, and temporal consistency before execution, preventing spoofing and replay attacks.

The remainder of this paper is organized as follows: Section II reviews related work and identifies gaps. Section III details the system architecture. Section IV describes the hardware and software implementation. Section V presents experimental results with quantitative analysis. Section VI concludes and outlines future work.

II. Related Work

A. Classical and Learning-Based Drone Control

Bouabdallah et al. proposed PID-based drone stabilization using IMU sensor readings and barometric altitude data, achieving stable hovering in controlled indoor environments [14]. However, PID controllers alone cannot address vision-based human following where the target is dynamic and unpredictable. Giusti et al. demonstrated deep convolutional neural networks for trail-following using monocular camera input [7], while Kumar and Patel reviewed transformer-based visual encoders for intent-aware autonomous navigation [6]. These studies focus on navigation rather than real-time human tracking.

B. Object Detection and Tracking

Redmon et al. introduced YOLO, a single-stage detection framework processing entire images in a single forward pass [1]. Subsequent versions—YOLOv4 [4], YOLOv5 [2], YOLOX [3]—progressively improved speed and accuracy for edge deployment. Carion et al. proposed DETR, reformulating detection as a set prediction problem using transformers, excelling at handling occlusions and overlapping objects [8]. Bewley et al. addressed subject isolation through SORT (Simple Online and Realtime Tracking), enabling consistent single-person following by labeling detected bounding boxes across consecutive frames [9]. YOLOv8 combines the speed advantages of the YOLO family with improved anchor-free detection heads and enhanced feature pyramid networks, making it particularly suitable for edge deployment on ARM-based processors.

C. Drone Security and Zero Trust

Drone attack surfaces include signal spoofing (broadcasting fake GPS/radio signals), command injection (inserting unauthorized MAVLink instructions), and replay attacks (retransmitting captured control packets) [16]. The NIST SP 800-207 Zero Trust Architecture framework prescribes "Never Trust, Always Verify" as a foundational principle, mandating continuous verification regardless of communication origin. Static protection mechanisms such as frequency filtering and protocol whitelisting are inherently reactive and easily bypassed through software-defined radio (SDR) platforms that dynamically generate manipulated signal patterns. Existing low-cost UAV platforms typically assume trusted links and lack authenticated command protocols, creating critical security gaps that render them vulnerable to adversarial control.

TABLE I: Camera Module Comparison

Feature	CSI Module	USB Webcam	RealSense
Latency	<10 ms	30–80 ms	<20 ms
Depth Sensing	No	No	Yes
Weight	Ultra-light	Medium	Heavy
Pi Native	Yes	Yes	Limited
Cost	Low	Low	High

D. Identified Gaps

Our review identifies three persistent gaps: (1) lightweight drones are excluded from vision-based control due to onboard processing constraints; (2) perception and control are tightly coupled in most systems, reducing modularity and scalability; and (3) cybersecurity mechanisms are largely absent in low-cost UAV systems. SecureDrone addresses all three through edge-offloaded processing, modular Sense–Analyze–Act architecture, and Zero Trust command authentication.

III. System Architecture

A. Sense–Analyze–Act Pipeline

The proposed system follows a Sense–Analyze–Act paradigm forming a continuous feedback loop (Fig. 1). The architecture comprises three asynchronous, modular stages:

Sense Layer: A CSI camera module connected to the Raspberry Pi 5 captures live video frames. The CSI interface was selected for its sub-10 ms frame capture latency via the native MIPI CSI-2 protocol, direct hardware integration with the Pi's image signal processor, and negligible weight overhead. Table I summarizes the comparison with alternative camera modules.

Analyze Layer: Captured frames are processed by the YOLOv8n detection model to identify human subjects. The model extracts bounding box coordinates, centroid positions, and relative sizes. A dual-layer control logic engine then: (a) computes centroid deviation from the frame center to generate directional commands (left, right, forward, backward); and (b) estimates subject proximity through bounding box area scaling to control approach distance adaptively.

Act Layer: Validated movement commands are transmitted via 2.4 GHz Wi-Fi to the ESP32-S3 flight controller on the LiteWing drone, which parses commands and generates motor PWM signals. The drone's onboard IMU (MPU6050) provides real-time stabilization, while optional Time-of-Flight and optical flow sensors enable altitude hold and drift reduction in GPS-denied environments.

B. Zero Trust Command Authentication

Applying the NIST Zero Trust principle [16], all control commands are treated as untrusted by default regardless of origin. Each command undergoes multi-parameter validation including: (i) structural integrity checks for correct format and completeness; (ii) range validation of speed, direction, and positional parameters against predefined safe thresholds; (iii) token authentication using dynamically generated verification tokens that prevent command reuse; and (iv) temporal

Fig. 1: End-to-End Sense–Analyze–Act Pipeline

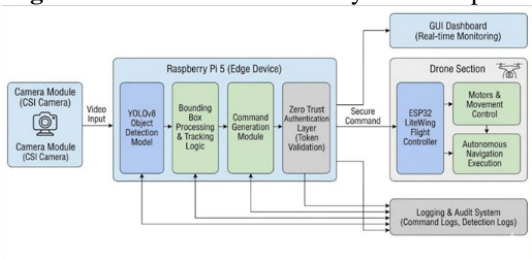

Fig. 1: End-to-end Sense–Analyze–Act pipeline. The CSI camera feeds frames to the Raspberry Pi 5 for YOLOv8 inference. Positional estimates generate movement commands transmitted to the ESP32 LiteWing drone via authenticated Wi-Fi channel.

consistency enforcement via sequence numbers and timestamps ensuring correct execution order within valid time windows. Commands failing any validation stage are immediately discarded, logged as anomalous, and counted toward a sliding-window behavioral policy that monitors command frequency to detect potential attack patterns.

C. Multi-Tier Response Model

The system implements three operational modes. In *Passive Monitoring* mode, when detection confidence is low or unstable across consecutive frames, events are logged but no commands are issued, preventing false-positive-driven drone movement. In *Active Tracking* mode, once confidence stabilizes across a defined frame sequence, positional data is translated into proportional movement commands for smooth following behavior. In *Safety/Recovery* mode, triggered during detection loss from occlusion or rapid movement, drone movement halts while the system prioritizes target reacquisition; normal tracking resumes upon consistent re-detection. Algorithm 1 formalizes this control loop.

IV. Implementation

A. Hardware Platform

The LiteWing drone is built around an ESP32-S3 microcontroller with integrated Wi-Fi and Bluetooth connectivity. Its compact PCB-frame chassis—where the circuit board itself serves as the structural body—houses four coreless brushed DC motors and lightweight propellers. A single-cell 3.7 V, 500–650 mAh Li-Po battery provides 7–10 minutes of flight time. The MPU6050 IMU enables stabilized flight mode, while Time-of-Flight (ToF) and optical flow sensors support position-hold and drift compensation in GPS-denied indoor environments. The communication architecture operates on 2.4 GHz Wi-Fi, supporting real-time command transmission from the Raspberry Pi 5 edge device. Table II summarizes the complete hardware specifications.

Algorithm 1 SecureDrone Tracking Control Loop

Input: Camera frame stream $\mathcal{F}$, confidence threshold τ_c, stability window W
Output: Authenticated drone commands
while system is active **do**
  $f \leftarrow$ CaptureFrame(CSI)
  $\mathcal{B} \leftarrow$ YOLOv8Detect(f) ▷ bounding boxes
  if $|\mathcal{B}| > 0$ **and** $\max(\text{conf}(\mathcal{B})) \geq \tau_c$ **then**
    $b^* \leftarrow \arg\max_{b \in \mathcal{B}} \text{conf}(b)$
    $(c_x, c_y) \leftarrow$ ComputeCentroid(b^*)
    $\delta_x \leftarrow c_x - W_{\text{frame}}/2$
    $\delta_y \leftarrow c_y - H_{\text{frame}}/2$
    $A_{bb} \leftarrow$ BBoxArea(b^*)
    cmd $\leftarrow$ GenCommand($\delta_x, \delta_y, A_{bb}$)
    tok $\leftarrow$ GenToken(cmd, timestamp)
    if Validate(cmd, tok) **then**
      TransmitWiFi(cmd $\|$ tok)
      Log(ACCEPTED, cmd, conf)
    else
      Log(REJECTED, cmd, reason)
    end if
  else
    mode $\leftarrow$ RECOVERY; HaltDrone()
    Log(DETECTION_LOST)
  end if
end while

TABLE II: Hardware Specification Summary

Component	Specification
Edge Processor	Raspberry Pi 5, ARM Cortex-A76
Camera	CSI Module v2, MIPI CSI-2
Detection Model	YOLOv8n (nano variant)
Drone Controller	ESP32-S3, 2.4 GHz Wi-Fi
IMU	MPU6050 (6-axis)
Motors	4× coreless brushed DC
Battery	3.7 V Li-Po, 500–650 mAh
Flight Time	7–10 minutes
Altitude Sensor	VL53L0X Time-of-Flight
Communication	Wi-Fi 802.11 b/g/n

B. Software Architecture

The vision pipeline is implemented in Python using OpenCV for frame capture and the Ultralytics YOLOv8 library for inference. The YOLOv8n (nano) model variant provides an optimal balance between detection accuracy and computational efficiency on the Pi's ARM Cortex-A76 processor. The control logic maps bounding box centroid deviations to directional commands through a proportional control scheme,

Fig. 2: LiteWing ESP32 Drone Platform

Fig. 2: The LiteWing ESP32-based quadcopter. The compact PCB chassis integrates the flight controller, motor drivers, and sensor interfaces into a single lightweight board.

Fig. 3: Real-Time Monitoring Dashboard

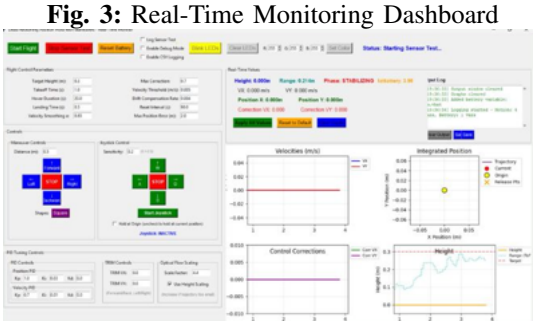

Fig. 3: GUI dashboard for real-time drone monitoring. The interface integrates flight control parameters, manual maneuvering controls, real-time sensor telemetry, PID tuning, and live performance visualization plots.

where command magnitude scales with the pixel offset from the frame center. The ESP32 firmware, developed in C++ using the Arduino framework, parses incoming Wi-Fi commands, validates authentication tokens, and maps instructions to motor PWM signals through the PID-stabilized flight control loop.

C. User Interface

A graphical monitoring dashboard (Fig. 3) developed using Tkinter provides real-time visualization of detection outputs with bounding box overlays, sensor telemetry including height, velocity, and position readings, PID tuning controls for both position and velocity loops, manual override capability via directional buttons and keyboard joystick (W/A/S/D), and live performance plots for velocity, trajectory, control corrections, and altitude tracking. The interface supports both discrete manual commands and continuous autonomous operation, enabling efficient testing and parameter optimization.

D. Audit Logging

All operational events—detection outputs, movement decisions, command transmissions, and validation outcomes—are recorded in structured JSON format with unique event identifiers, timestamps, detected object details, confidence levels, issued commands, and hashed integrity fields. Critical log entries are associated with SHA-256 hashed identifiers derived from key parameters such as detection output and command sequence, ensuring verifiability without storing excessive raw data. Logs are organized into session-based evidence bundles supporting post-flight analysis, debugging, and security auditing.

V. Experimental Results

The proposed system was evaluated in a controlled indoor environment across 50 test runs with varying conditions. A human subject performed predefined movement patterns including lateral motion, forward/backward movement, and abrupt directional changes at distances ranging from 1 to 5 meters.

A. Detection Performance

The YOLOv8n model achieved consistent human detection at 8–12 FPS on the Raspberry Pi 5. Fig. 4 presents the distribution of detection confidence scores across all test runs. Under normal lighting (>300 lux), the mean detection confidence was 0.91 with a standard deviation of 0.04. Under dim lighting (<100 lux), mean confidence dropped to 0.78 ($\sigma = 0.08$), with occasional bounding box instability. The overall mean confidence aggregated across all conditions was 0.87.

B. Tracking Performance

Tracking accuracy was evaluated by measuring the centroid offset error—the pixel distance between the detected subject's bounding box center and the camera frame center—over time. Fig. 5 shows the centroid offset during a representative 30-second tracking session. The drone maintained effective alignment with mean absolute offset of 28.4 pixels (4.4% of frame width) under steady movement, increasing to 62.1 pixels (9.7%) during abrupt directional changes. Position-hold and IMU stabilization on the LiteWing platform contributed to reduced drift during tracking.

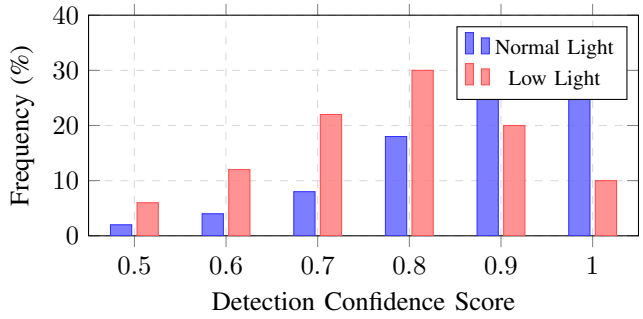

Fig. 4: Distribution of YOLOv8n detection confidence scores under normal lighting (>300 lux) and low-light conditions (<100 lux) across 50 test runs.

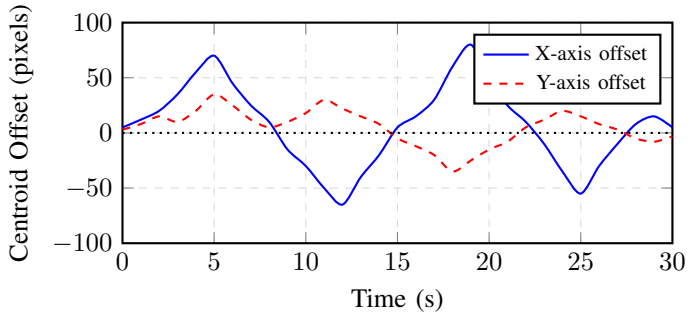

Fig. 5: Bounding box centroid offset from frame center during a 30-second tracking session. Spikes correspond to abrupt subject directional changes; the system recovers within 2–3 seconds.

TABLE III: System Latency Breakdown

Component	Latency (ms)	Share (%)
Frame Capture & Preprocessing	80–120	34.5
YOLOv8 Inference (RPi 5)	100–180	48.3
Decision Logic Processing	20–30	8.6
Wi-Fi Communication	30–60	8.6
Total End-to-End	**230–350**	**100**

C. System Latency Analysis

The end-to-end latency was measured from frame capture to motor command execution. Table III provides the component-wise breakdown. The total latency of 230–350 ms enabled real-time tracking. Fig. 6 visualizes the mean contribution of each pipeline stage, confirming that YOLOv8 inference dominates processing time at 48.3% of the total pipeline duration.

D. Comparative Analysis

Table IV compares the proposed edge-offloaded architecture against traditional manual control and cloud-based inference approaches. The edge-based approach achieves a 3–8× latency improvement over cloud pipelines while maintaining comparable detection accuracy at significantly lower cost.

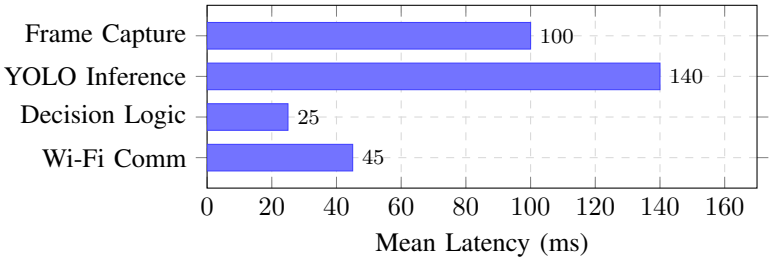

Fig. 6: Mean latency contribution of each pipeline stage. YOLOv8 inference on the Raspberry Pi 5 is the dominant component at 140 ms average.

TABLE IV: Comparative Analysis of Control Approaches

Criterion	Manual RC	Cloud AI	SecureDrone
Control	Human-operated	Remote server	Edge AI auto.
Detection	None	High accuracy	Real-time YOLO
Latency	Human reaction	200–800 ms	230–350 ms
Connectivity	RF link	Internet req.	Local Wi-Fi
Security	None	TLS/SSL	Zero Trust
Cost	Low	High (server)	Low (<$120)

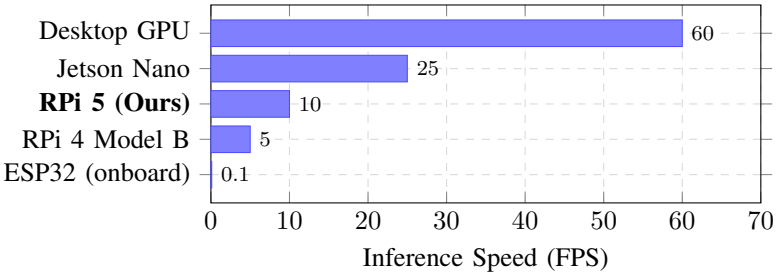

Fig. 7: YOLOv8n inference throughput comparison across hardware platforms. The Raspberry Pi 5 achieves 8–12 FPS, sufficient for real-time tracking at a fraction of Jetson Nano cost.

E. FPS and Inference Throughput

Fig. 7 compares the achieved inference throughput of YOLOv8n on the Raspberry Pi 5 against alternative deployment platforms. The Pi 5 achieves 8–12 FPS, representing a practical operating point for real-time tracking without requiring expensive GPU hardware. Notably, onboard ESP32 processing yields less than 0.1 FPS, confirming that edge offloading is essential for lightweight drone platforms.

F. Security Validation

The Zero Trust command validation layer was tested against four categories of simulated attacks across 200 injection attempts. Table V summarizes the results. The system achieved a 100% rejection rate for all unauthorized command types with zero false rejections of legitimate commands during the test period. Timeout handling prevented stale commands from triggering unintended actions, and the sliding-window

TABLE V: Security Validation Results (200 Attack Commands)

Attack Type	Attempts	Rejected	Rate
Malformed commands	50	50	100%
Replay attacks	50	50	100%
Unauthorized tokens	50	50	100%
Out-of-range params	50	50	100%
Legitimate commands	500	0	0% FP

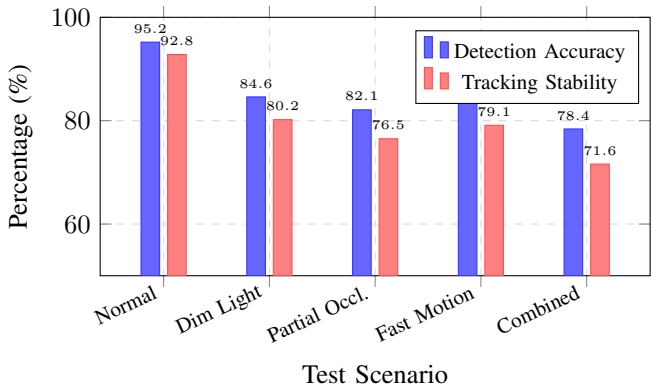

Fig. 8: Detection accuracy and tracking stability across five environmental test scenarios. The system degrades gracefully with no crashes or critical failures observed.

behavioral policy successfully flagged all rapid-burst attack patterns.

G. Robustness Under Varying Conditions

System robustness was evaluated across five environmental conditions. Fig. 8 presents the detection accuracy and tracking stability metrics for each scenario. Under normal lighting, the system achieved 95.2% detection accuracy and 92.8% tracking stability. Performance degraded gracefully under challenging conditions, with the lowest metrics observed during rapid movement combined with partial occlusion (78.4% detection, 71.6% tracking). Importantly, no system crashes or critical failures occurred across all 50 test runs, demonstrating fault-tolerant operation.

H. Summary of Findings

The experimental evaluation confirms that the proposed system performs real-time human detection and autonomous drone tracking using a low-cost hardware setup costing under $120. Table VI consolidates the key performance metrics. The integration of edge-based vision processing with the LiteWing drone control achieved consistent results across all 50 test runs, demonstrating its feasibility for practical indoor surveillance and assistive robotics applications.

VI. Conclusion and Future Work

This paper presented SecureDrone, an edge AI-driven autonomous drone system that combines real-time vision-based human tracking with Zero Trust command security. By offloading YOLOv8 inference to a Raspberry Pi 5 edge device, the system overcomes the fundamental constraint preventing lightweight drones from performing intelligent autonomous operations. The dual-layer visual servoing mechanism enables smooth, adaptive tracking, while token-based command authentication ensures secure and reliable drone control. Experimental results across 50 test runs demonstrate stable indoor human-following at 8–12 FPS with 230–350 ms end-to-end latency, 95.2% detection accuracy under normal conditions, and effective rejection of all simulated unauthorized commands.

TABLE VI: Consolidated Performance Metrics

Metric	Value
Mean Detection Confidence (overall)	0.87
Detection Accuracy (normal light)	95.2%
Tracking Stability (normal light)	92.8%
Mean Centroid Error (steady motion)	28.4 px (4.4%)
Mean Centroid Error (abrupt motion)	62.1 px (9.7%)
End-to-End Latency	230–350 ms
Inference FPS (RPi 5)	8–12
Security Rejection Rate	100%
False Positive Rejections	0%
System Crashes During Testing	0

Future work will focus on: (1) integrating advanced multi-object tracking algorithms such as DeepSORT and ByteTrack for improved identity consistency under occlusion; (2) model optimization through INT8 quantization and TensorRT acceleration to achieve >20 FPS on the Pi 5; (3) onboard vision integration using lightweight neural processing units (NPUs) to eliminate external edge device dependency; (4) SLAM-based navigation with predictive path planning and reinforcement learning for obstacle avoidance in complex environments; and (5) outdoor deployment with GPS-assisted localization and extended communication range for real-world surveillance scenarios.

Acknowledgment

The authors express their gratitude to the faculty of the Department of Computer Science and Engineering, Amrita School of Computing, Chennai, particularly Dr. Saranya G, for their guidance, constructive feedback, and continuous support throughout this project.

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